

Ch1 Linear Algebra: Directions, Projections, Cosine

Source: reading_pack/md/A01_11.md | Full textbook: TEXTBOOK_Stability_Identifiability.md

1.1

1.1

Let $\theta \in \mathbb{R}^D$, $\|\theta\|=1$. For a vector $x \in \mathbb{R}^D$, the cosine of the angle between x and θ is

$$\text{score} = \theta^T x = \|x\| \cdot \cos(x, \theta)$$

For a vector $x \in \mathbb{R}^D$, the cosine of the angle between x and θ is $\cos(x, \theta) = \frac{\theta^T x}{\|x\|}$. Park (2019) sparse linear combination $\sum c_i f_i$ where $f_i = \theta_i$.

1

$$\cos(\theta_i, \theta_j) = \frac{\theta_i^T \theta_j}{\|\theta_i\| \|\theta_j\|} \in [-1, 1]$$

• $\theta_i^T \theta_j \in [-1, 1]$

• $\text{Stab_boot} = E[\cos(\theta_i, \theta_j)]$ 20 bootstrap

2

Let $w \in \mathbb{R}^D$, $\theta = w/\|w\|$. Then $\alpha \sigma \theta = \alpha \theta$. The projection of x onto θ is $P_\theta(x) = (\theta^T x) \theta$.

3 Ch6

LoRA $W + \Delta W$, $\Delta W = b \cdot a^T$. Wang (2019) $W + W \cdot b \cdot a^T$. Wang (2019) $W \cdot b$ head

**Elhage p.1-6 Park p.4-6 (2)

1.2

• $\theta_i^T \theta_j \in [-1, 1]$, $E[\cos] = 0$, $\text{Var} \approx 1/D$. $D=896$, $\cos \approx 0.03$ to 0.979 .

• $\mu_i = [1, 0]$, $\mu_j = [-1, 0]$, $\theta = \mu_i - \mu_j = [2, 0]$, $\theta^T = [1, 0]$, $|T|$ mass-mean
